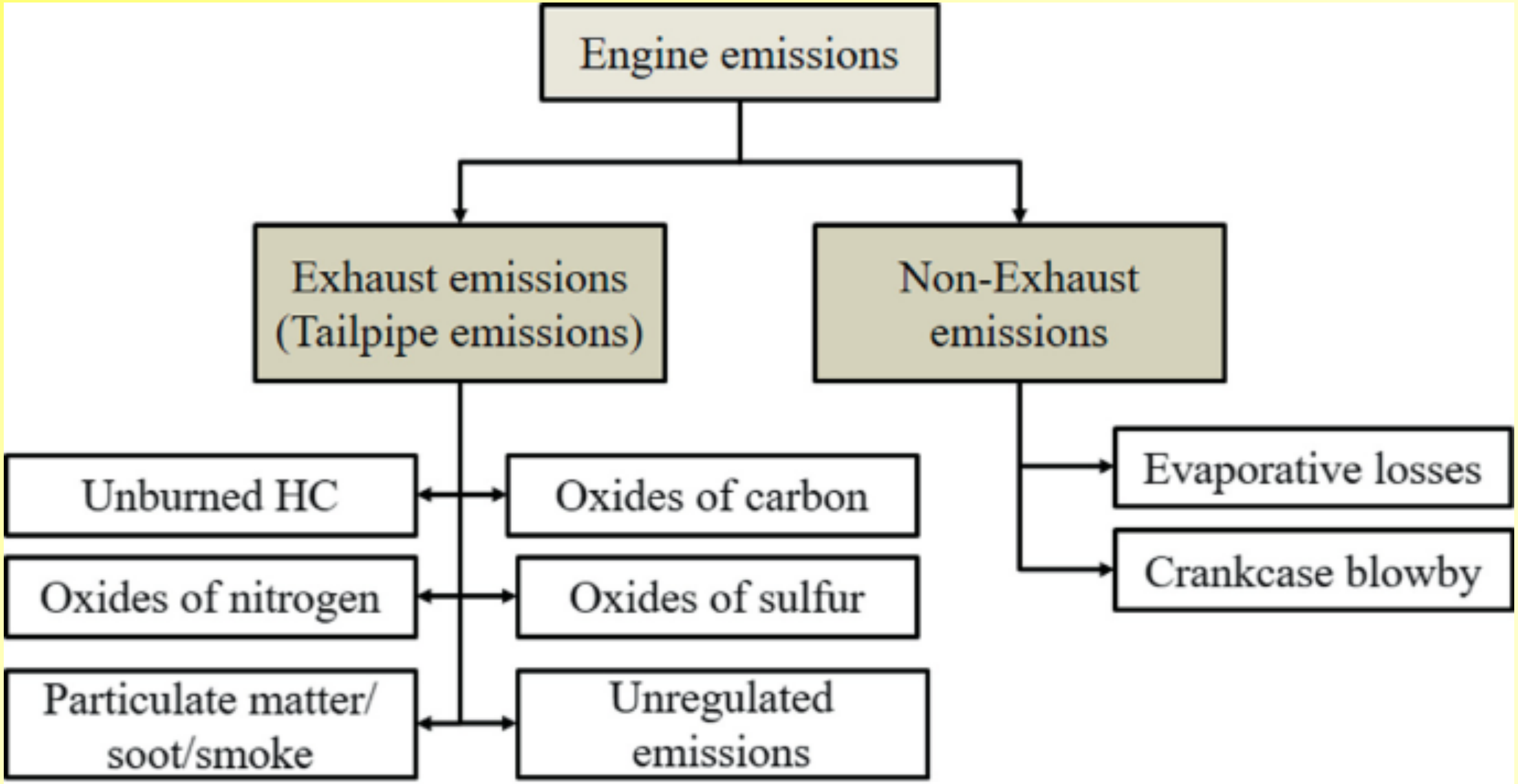
AIR POLLUTION



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SYLLABUS

S.I. Engines: Deviation from the ideal cycle, combustion losses, flame development and propagation, effects of operating variable on detonation; Detonation rating of engine fuels, HUCR Dope, preignition combustion chambers, Mechanics of carburetion, MPFI systems, mixture requirement and combustion.

C.I. Engines: Stages of combustion, Diesel knock, effect of operating variables, knock rating of C.I. engine fuels, Cetane number; Effect of additives, combustion chambers. Requirements of fuel injection system and its mechanism, Fuel filters.

Lubrication and Cooling: Forced, splash, by-pass and direct, Blow by and crankcase dilution.
Air and water cooling, Thermostatic control.

Performance and Testing: High speed indicators, various efficiencies, Morse Test and Willan's line; torque and mean effective pressure, performance and heat balance sheet, Measurement of volumetric efficiency, Effect of atmospheric condition on performance of I.C. Engines; Supercharging; high altitude problem of I.C. Engines, methods and types of supercharging; suitability of S.I. and C.I. Engines;

Air Pollution: Vehicular emission, its norms, Causes and control of air pollution from diesel and petrol engines, catalytic converter.
Introduction to dual fuel and multi-fuel engines; free-piston engine and rotary combustion engine.

INTRODUCTION

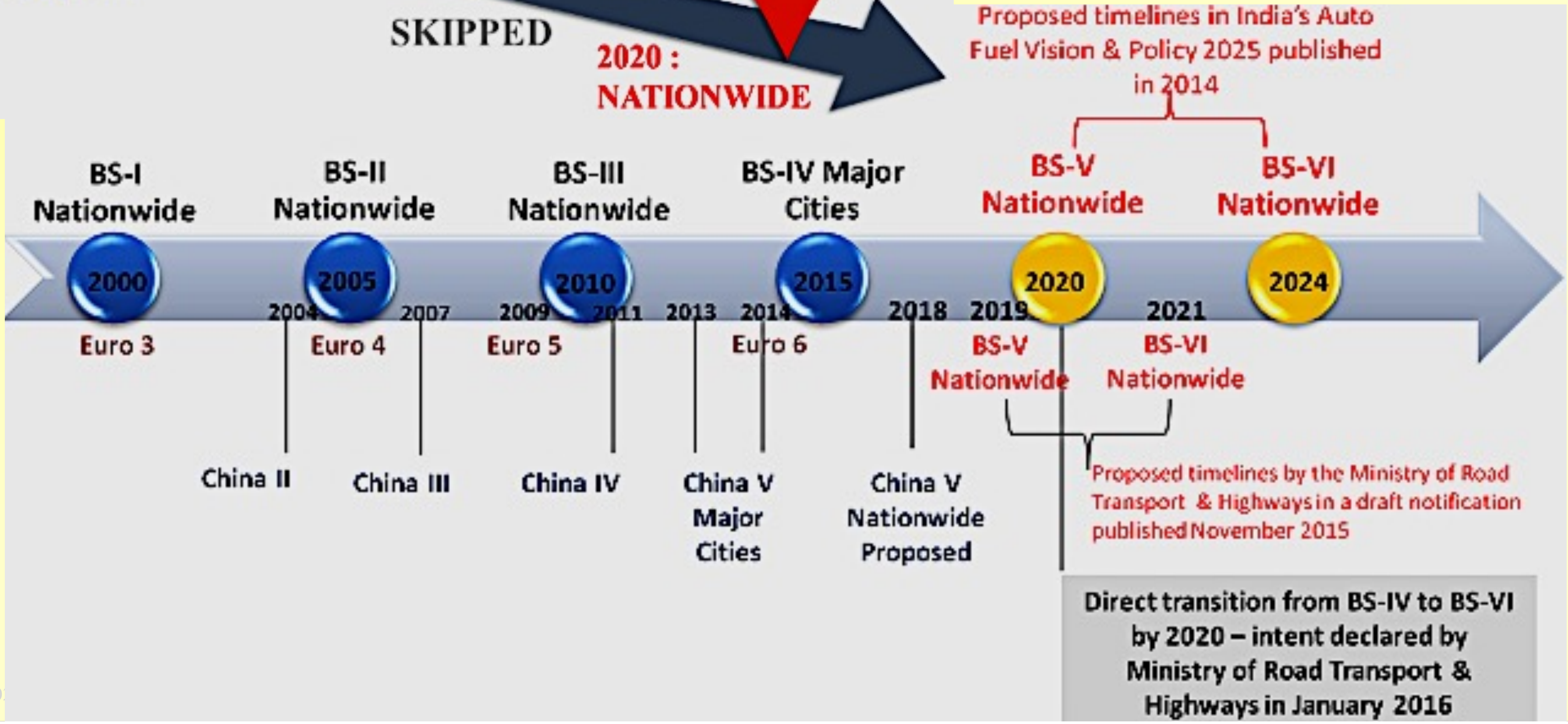
- ***The Vehicular Emissions (unburnt HCs, CO_2, CO, NO_x, SO_x, solid carbon particulates etc.) exhausted into surroundings, pollute atmosphere and causes the following problems:***
  - ✓ 1. Global warming
  - ✓ 2. Smog
  - ✓ 3. Odour
  - ✓ 4. Acid rain
  - ✓ 5. Respiratory and other health hazards
- ***So, it is required to develop Engines and Fuels such that they generate very few quantity of***
  - ✓ Harmful emissions and ***Left*** over in surroundings
  - ✓ Without major / less ***impac***t on environment
- ***In present scenario it is not possible, so***
  - ✓ ***After treatment*** of exhaust gases and
  - ✓ ***In-cylinder*** reduction of emissions are important
- ***In case of After Treatment, it includes***
  - ✓ The use of ***thermal*** or ***catalytic converters*** and
  - ✓ Particulate ***traps***
- ***For in-cylinder reductions, the***
  - ✓ Exhaust gas ***recirculation*** and
  - ✓ Some fuel ***additives*** are used

AUTOMOTIVE AIR POLLUTION (J B Heywood)

Pollutant	Impact	Mobile source emissions as % of total†	Automobile emissions: Uncontrolled vehicles, g/km‡	Automobile emissions: Reduction in new vehicles, % ¶	Truck emissions††: SI engines, g/km	Truck emissions††: Diesel, g/km
Oxides of nitrogen (NO and NO_2)	Reactant in photochemical smog; NO_2 is toxic	40–60	2.5	75	7	12
Carbon monoxide (CO)	Toxic	90	65	95	150	17
Unburned hydrocarbons (HC, many hydrocarbon compounds)	Reactant in photochemical smog	30–50	10	90	17‡‡	3
Particulates (soot and absorbed hydrocarbon compounds)	Reduces visibility; some of HC compounds mutagenic	50	0.5§	40§	n	0.5

EMISSION CONTROL : TIMELINE TRAIL

Euro testing is done at sub-zero temp & 120 km/h while Bharat is done at 24-28^{o}C & 90 km/h.



EUROPEAN EMISSION STANDARDS FOR CARS *(g/km)*

Tier	Date	CO	THC	NMHC	NO_x	$HC+NO_x$	PM	P***
Diesel								
Euro 1†	July 1992	2.72 (3.16)	-	-	-	0.97 (1.13)	0.14 (0.18)	-
Euro 2	January 1996	1.0	-	-	-	0.7	0.08	-
Euro 3	January 2000	0.64	-	-	0.50	0.56	0.05	-
Euro 4	January 2005	0.50	-	-	0.25	0.30	0.025	-
Euro 5	September 2009	0.50	-	-	0.180	0.230	0.005	-
Euro 6 (future)	September 2014	0.50	-	-	0.080	0.170	0.005	-
Petrol (Gasoline)								
Euro 1†	July 1992	2.72 (3.16)	-	-	-	0.97 (1.13)	-	-
Euro 2	January 1996	2.2	-	-	-	0.5	-	-
Euro 3	January 2000	2.3	0.20	-	0.15	-	-	-
Euro 4	January 2005	1.0	0.10	-	0.08	-	-	-
Euro 5	September 2009	1.0	0.10	0.068	0.060	-	0.005**	-
Euro 6 (future)	September 2014	1.0	0.10	0.068	0.060	-	0.005**	-

* Before Euro 5, passenger vehicles > 2500 kg were type approved as light commercial vehicles N_1-I

** Applies only to vehicles with direct injection engines

*** A number standard is to be defined as soon as possible and at the latest upon entry into force of Euro 6

† Values in brackets are conformity of production (COP) limits

BHARAT EMISSION LIMITS FOR LIGHT MOTOR VEHICLES

Compression Ignition	CO	HC	HC + NOx	NOx	PM	PN
BS IV	0.50 – 0.74	–	0.30 – 0.46	0.25 – 0.39	0.025 – 0.06	–
BS VI	0.50 – 0.74	–	0.17 – 0.215	0.08 – 0.125	0.0045	6×10^{11}

Positive Ignition	CO	HC	HC + NOx	NOx	PM	PN
BS IV	1.0 – 2.27	0.1 – 0.16	–	0.08 – 0.11	–	–
BS VI	1.0 – 2.27	0.10 – 0.16	–	0.060 – 0.082	0.0045	6×10^{11}

Norms	European	Year	CO (g/Km)	HC + NO_x (g/Km)
1991 Norms	–	–	14.3–27.1	2.0(Only HC)
1996 Norms	–	–	8.68–12.40	3.00–4.36
1998 Norms	–	–	4.34–6.20	1.50–2.18
India Stage 2000 Norms	Euro 1	2000	2.72	0.97
Bharat Stage-II	Euro 2	2001	2.2	0.50
Bharat Stage-III	Euro 3	2005	2.3	0.35
Bharat Stage-IV	Euro 4	2010	1.0	0.18
Bharat Stage-V	Euro 5	2017	0.63	0.10
Bharat Stage-VI	Euro 6	2020	0.50	0.07

PETROL & DIESEL ENGINE EMISSION NOZRMS

➢ *Petrol emission*

Norm	CO (g/km)	HC (g/km)	NOx (g/km)	HC + NOx (g/km)	PM (g/km)
BS IV	1	0.1	0.08	-	Not specified
BS VI	1	0.1	0.06 (25%↓)	-	0.005

➢ *Diesel emission*

Norm	CO (g/km)	HC (g/km)	NOx (g/km)	HC + NOx (g/km)	PM (g/km)
BS IV	0.50	-	0.25	0.30	0.025
BS VI	0.50	-	0.08 (68%↓)	0.17 (43%↓)	0.0045 (82%↓)

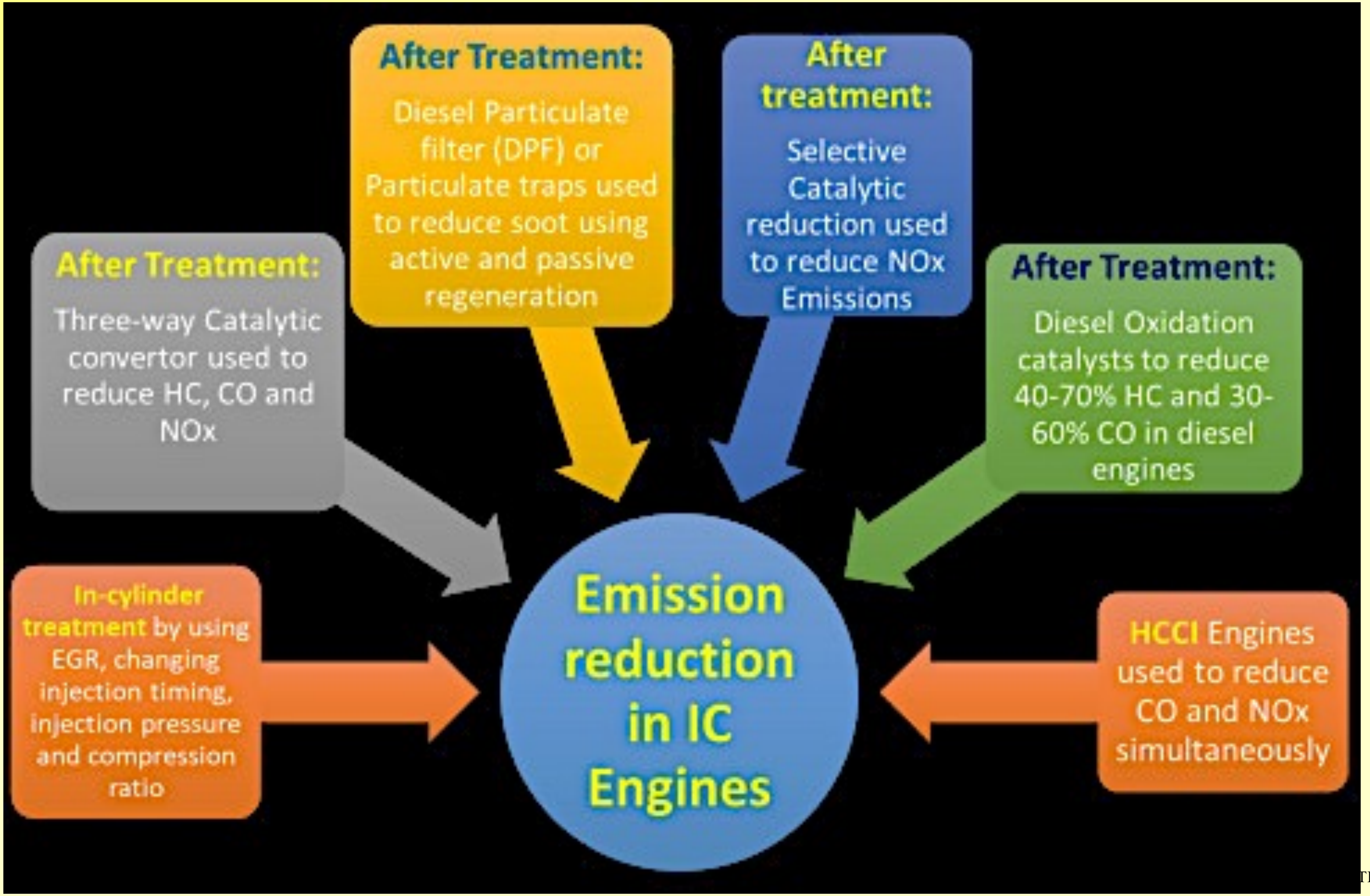
Source: Economics times   *For light Duty Vehicles

Sl. No.	Parameters	Unit	BS -VI Spec (Manufacturing)
1	Density @ 15^0 C	Kg/m3	821.5-845
2	Distillation- T -95	^{0}C max	360
3	Sulphur, Total	mg/kg max	8
4	Cetane Number	Min.	51.4
5	Cetane Index	Min.	46
6	Flash Point	^{0}C min	42
7	Kinematic Viscosity at 40^0C	Cst	2.15-4.5
8	PAH	% wt-max	11
9	Total Contaminants	mg/kg-max	24
10	Oxidation Stability	g/m3-max	21/18
11	RCR on 10% Residue	% wt-max	0.3
12	CCR on 10% Residue	% wt-max	-
13	Water Content	mg/kg-max	200
14	Lubricity corrected WSD	microns-max	420
15	Ash	%wt-max	0.01

1	Density @ 15^0 C	Kg/m3	720-773.7
2	Distillation		
	E-70	% Vol	11-45
	E-100	% Vol	40-70
	E-150	% Vol	75
	FBP	^{0}C max	200
	Residue	% Vol. Max	2
3	Sulphur, Total	mg/kg max	8
4	RON	Min.	91.4 (Note-1)
5	MON	Min.	81.4
6	RVP @ 38 deg C	Kpa	60
7	VLI (10RVP+7E70)		
	Summer (May to Jul)	Max	750
	Others	Max	950
8	Benzene	% Vol-max	1
9	Aromatics	% Vol-max	35
10	Olefin	% Vol-max	21
11	Existent Gum	g/m3-max	
12	Gum(Solvent washed)	mg/100 ml max	5
13	Oxidation Stability	Minutes-Min	360
14	Lead as Pb	g/l-max	0.005
15	Oxygen	%wt-max	2.7

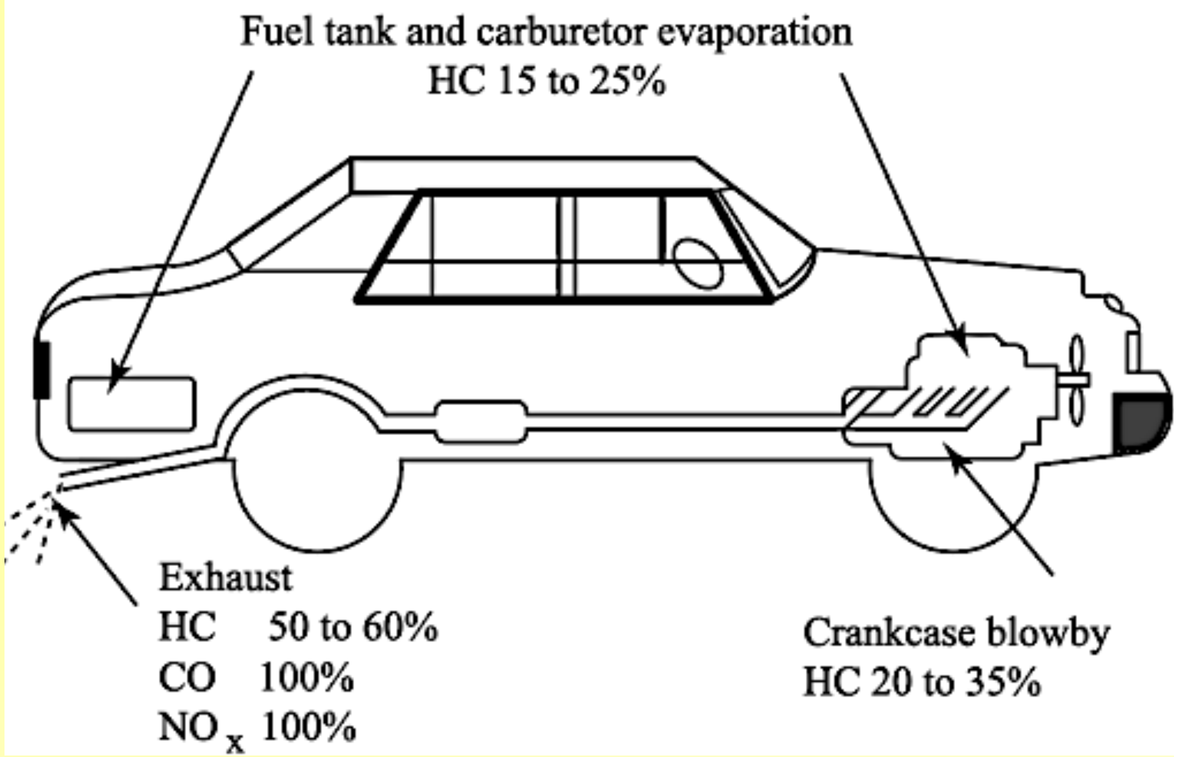
VEHICULAR EMISSIONS

- ***The statutory regulations set a limit of pollutants (NO_x, SO_x, CO_2, CO, particulate matter, HCs etc.) released by auto-mobiles, industries, power plants, diesel generators etc. into environment.***



VEHICULAR EMISSIONS

- ***Engine emissions***
  - 1. Exhaust emissions
    - *i. Unburnt hydrocarbons (HCs)*
    - *ii. Oxides of carbon (CO and CO_2)*
    - *iii. Oxides of nitrogen (NO and NO_2)*
    - *iv. Oxides of sulphur (SO_2 and SO_3)*
    - *v. Particulates matter/soot and smoke*
    - *First 4-common in SI-CI, last 2 in CIs*
  - 2. Non-exhaust emissions
    - *Unburnt HCs: from fuel tank, blowby*



- ***1. Exhaust emissions:***
  - 1. ***Unburnt*** HC*s*: in SI upto 6000 ppm (~1- 1.5%) in which 40% unburnt & 60% partially reacted;
    - ***1. Incomplete combustion: i. Improper mixing, ii. Flame quenching***
    - ***2. Crevice volumes & flow: volume between piston rings, sealing grooves, cylinder walls; 3.5% of fuel***
    - ***3. Leakage past EV: A-F mixture forced in crevice volume around EV & seat leaks in E-manifold***
    - ***4. Valve overlap: small amount of fresh charge escape inevitable, worst condition occurs at low speed***
    - ***5. Deposit on walls: fuel drops on CCs during compression-combustion-desorbed, older engines***
    - ***6. Oil on CC walls etc. oil film absorbed and desorb the fuel particles depends on gas pressure***
  - HCs from CI: operates on fuel-lean equivalence ratio, have ~1/5 HCs of SI with same reasons.
  - HCs a strong function of air-fuel ratio, CC geometry, engine operating parameters & speed etc.

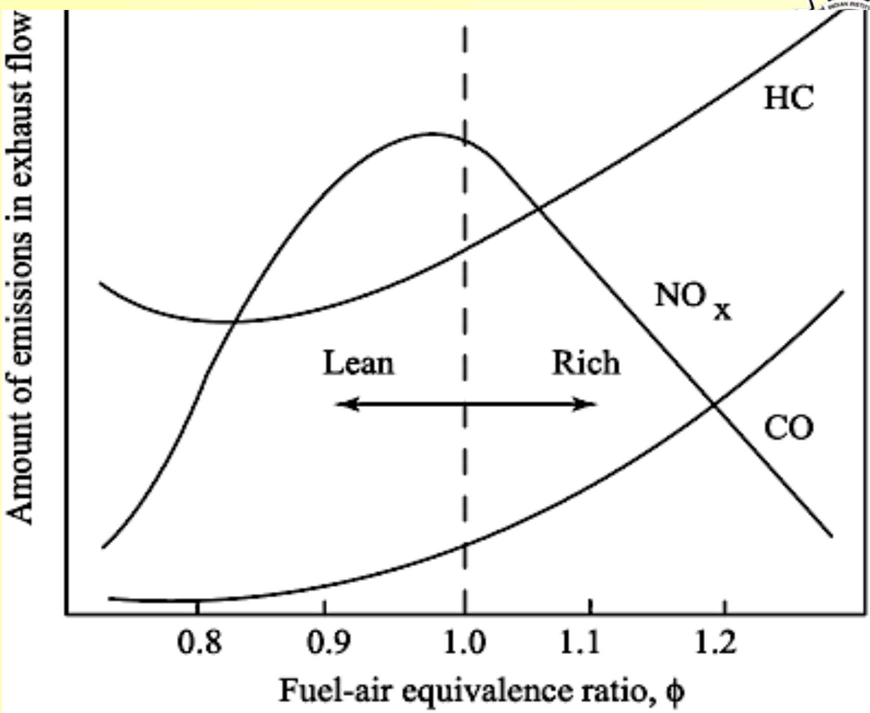
EXHAUST EMISSIONS

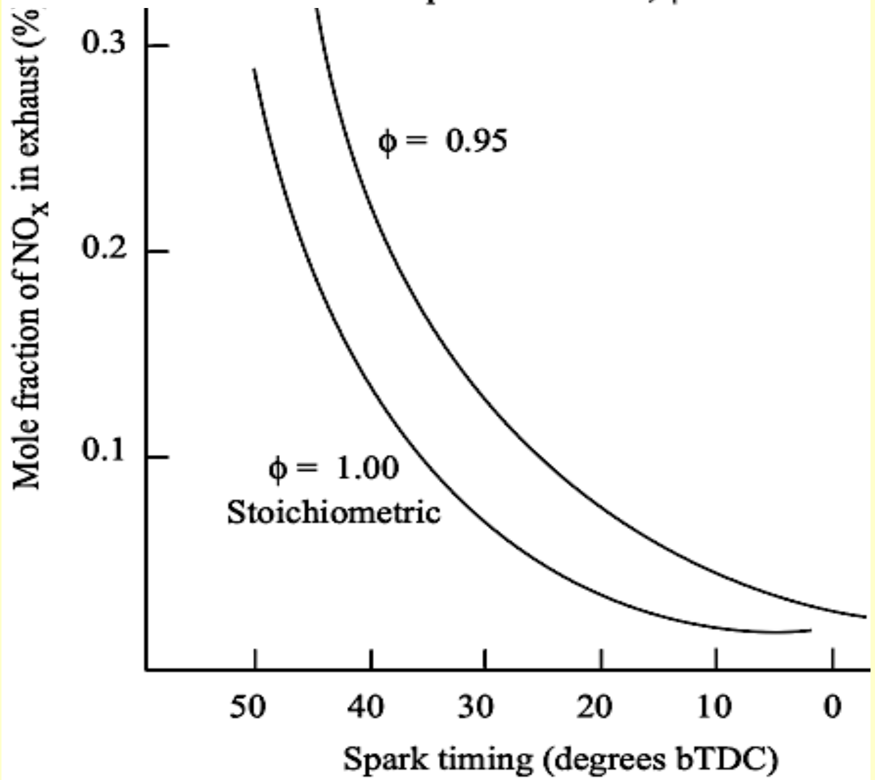
➢ *2. Carbon mono-Oxide emission & control*

- ✓ A colourless, odourless, poisonous gas generated with fuel rich equivalence ratio when
  - ***O_2 can't convert all carbon in CO_2 & ends up as CO***
- ✓ In SI exhaust, it is ~ 0.2 - 5%. Max CO is generated with rich mixture during starting, acceleration and overload.
- ✓ Poor mixing, rich regions & incomplete combustion be the source of CO; well design have CO fraction ~ 0.001.

➢ *3. Oxides of Nitrogen:*

- ✓ NO_x from N_2 in air combine with O_2, undesirable
- ✓ N_2 found in fuel-blend traces of NH_3, NC & HCN
- ✓ No_x formed to association of N_2 with O_2 at higher temp-pr. max NO_x formed with lean mixture.
- ✓ Exhaust have NO_x up to 2000 ppm, most is NO with small NO_2 & traces of NO_x form O_3 in atm.
- ✓ Photochemical smog:
  - ***NO_2 + sunlight → NO + O + smog, O+O_2 – O_3 . O_3***
  - ***It is harmful to lungs, tissues, plants-trees & crops***





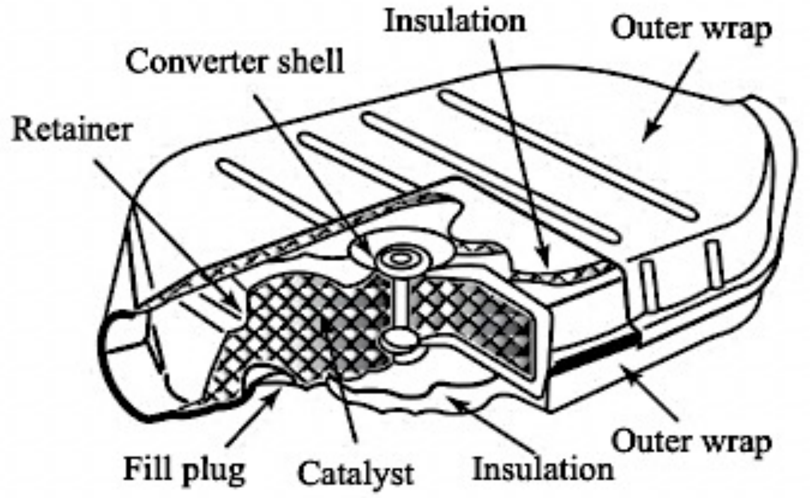
EMISSION : CAUSES & CONTROL

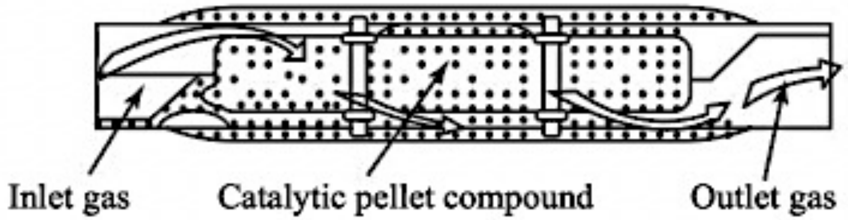
- ***4. Particulates Matter***
  - ✓ Exhaust of CI engines contain solid carbon spheres (9 nm to 90 nm, average size 15 - 30 nm,
  - ✓ A single particle contains upto 5000 carbon spheres) of HCs &
    - ***Traces of absorbed components are generated in the fuel rich zones during combustion.***

    $H_xC_y + zO_2 \rightarrow aCO_2 + bH_2O + cCO + dC(s)$
  - ✓ These are seen as exhaust smoke and cause an unbearable odours pollution.
  - ✓ Maximum density of particulate emissions occurs when engine is under load at WOT.
  - ✓ Upto 25% of carbon in soot come from lub oils &
    - ***Rest from fuel amounts ~ 0.2 – 0.5% of fuel***
- ***5. Other Emissions***
  - ✓ ***Aldehydes***: due to incomplete combustion in alcohol fuels, cause eye & respiratory irritant.
  - ✓ ***Sulphur***: CI engine fuels contains small amount of sulphur.
  - ✓ ***Unleaded*** gasoline contains ~ 150 to 550 ppm & fuels contain up to 5500 ppm by weight.
  - ✓ When exhausted as SO_x, contribute to ***acid rain*** problems
    - $H_2 + S \rightarrow H_2S,\ S + O_2 \rightarrow SO_2 + O_2 \rightarrow SO_3 + H_2O \rightarrow H_2SO_{3/4}$
  - ✓ ***Lead***: TEL (~0.15 g/liter) poisonous pollutant, hardens surfaces of CC, valves & seats.
  - ✓ 10 – 50% exhausted, remaining deposited on walls which reduces HCs emissions.
  - ✓ ***Phosphorus***: small amount in engine exhaust, come from fuel, lub oils and from atm air.

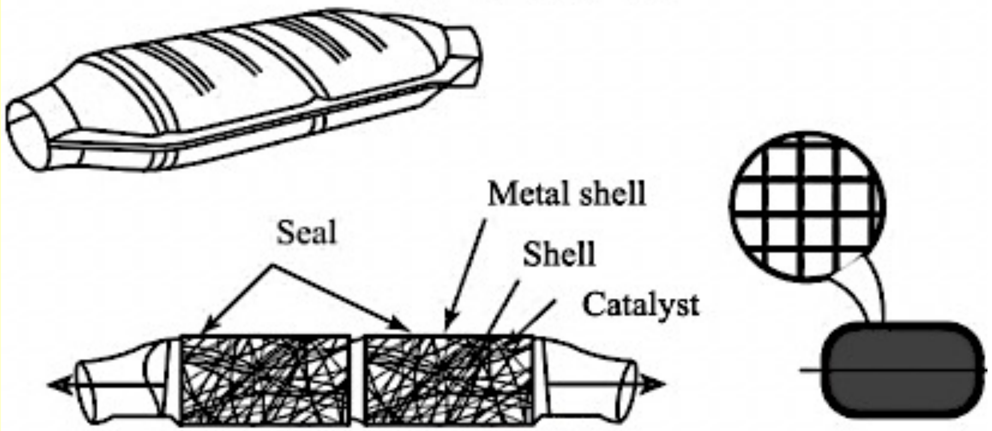
EMISSION : CAUSES & CONTROL

- ***For treatment, add-on devices in exhaust passage***
- ***1. Thermal converters for secondary reactions***
  - ✓ Work at high-temp (600 - 700^{o}C) with adequate retention time in exhaust manifolds to react O_2
  - ✓ $CO + O_2 \rightarrow CO_2$, $C_xH_x + zO_2 \rightarrow xCO_2 + H_2O$
  - ✓ Promote oxidation of CO & HCs, minimize heat losses & keep exhaust cool to lower emissions.
  - ✓ Thermal convertors doesn’t reduce NO_x emissions
- ***2. Catalytic converters***
  - ✓ Similar to thermal with certain catalyst materials to reduce secondary reaction temp to 200-300^{o}C.
  - ✓ Porous SS, ceramic honeycomb, loose granular Pd & Pt promotes oxidation reaction of CO/HCs
  - ✓ Rd promote NO_x reaction in exhaust flow passage to enhance, promote oxidation are most effective in reducing the engine emissions..
  - ✓ $NO + CO \rightarrow N_2 + CO_2$, $NO + H_2 \rightarrow N_2 + H_2O$.
  - ✓ In CI engines, alumina is used as ceramic material in convertors





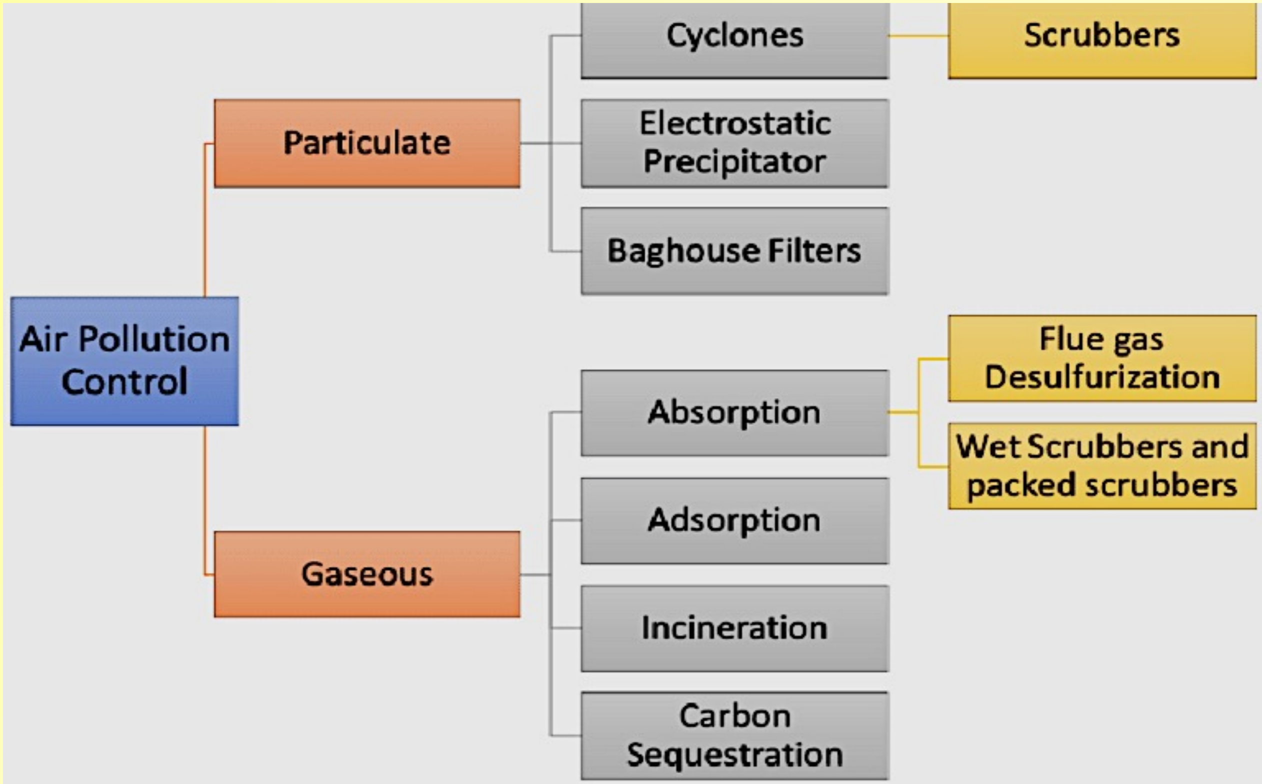
(a) Packed spheres

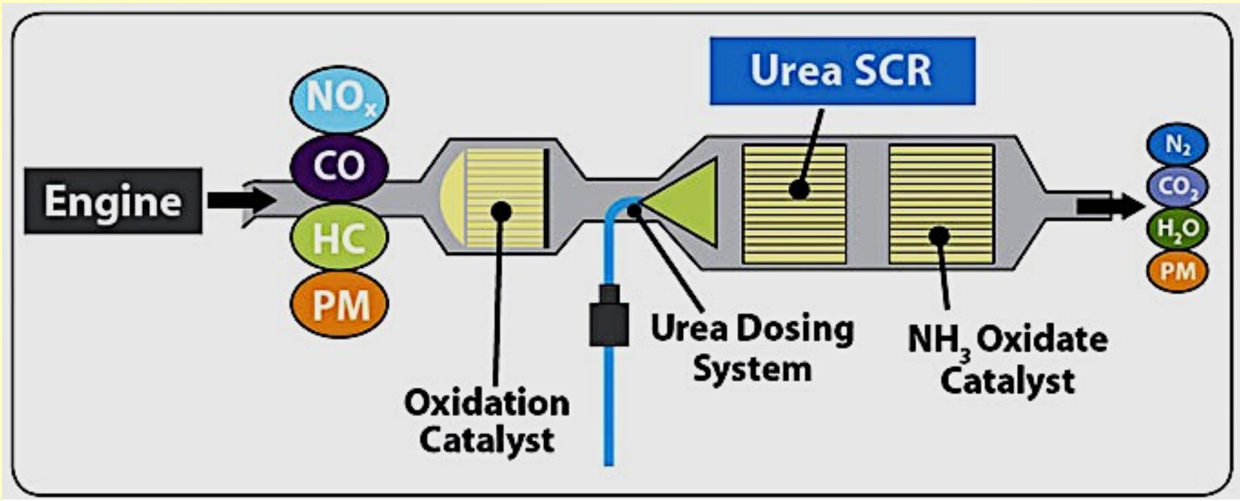


(b) Honey comb structure

EMISSION : CAUSES & CONTROL

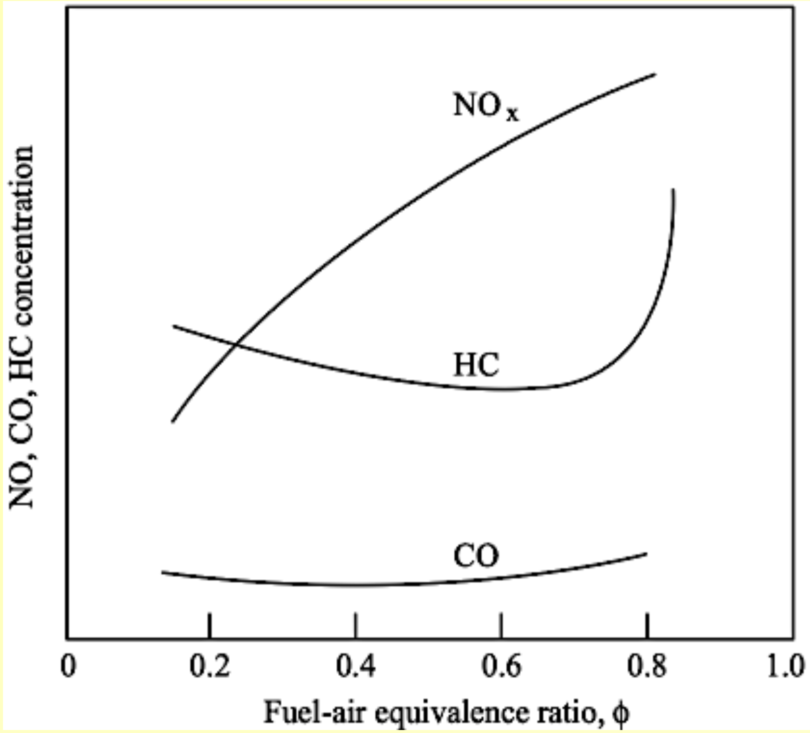
- ***Sulphur***
  - ✓ Pose problem for catalytic converter
  - ✓ Form SO_2 to SO_3 then converted to H_2SO_4 if exhaust temp reduces<400°C
  - ✓ SO_2 to SO_3 contribute to form H_2SO_4 causes the acid rain.
- ***Cold start-ups***
  - ✓ Catalytic converters inefficient in cold climates, 50% efficient in 250 - 300°C
  - ✓ Catalytic convertors preheating includes by,
  - ✓ 1. locating converter close to engine
  - ✓ 2. having super insulation
  - ✓ 3. employing electric preheating
  - ✓ 4. using flame heating and
  - ✓ 5. incorporating thermal batteries





CI EMISSION : CAUSES & CONTROL

- ***CI Engines: HC & CO reduced but can't NO_x due to low exhaust temp***
  - 1. Particulate traps:
    - ***Ceramic/metal wire-mesh filter, trap 60 - 90% particulates***
    - ***Carbon in muffler, ignited by spark @550-650°C to vacate***
    - ***Catalyst trap carbon soot ignited at reduced 350 to 450°C***
    - ***Re-generate by self igniting increase of back-pr due to soot***
    - ***It is simple, economical, reliable, reduce fuel consumption***
  - 2. Modern diesel engines reduce soot particulates by
    - ***Advanced design of fuel system***
    - ***Combustion chamber geometry***
    - ***Better injection, CC design and higher pressure,***
    - ***Hot spray promote turbulence/swirl mixing air fuel quicker***
  - 3. Emission NO_x reduction by Cyanuric acid (solid)
    - ***Exhaust dissociate, produce isocyanide that***
      - React with NO_x to form N_2, H_2O & CO_2 at ~500°C
      - Reduce NO_x up to 95% without loss in engine performance
    - ***Practically not possible in automobiles due to its size, weight & complexity***
    - ***Research is done using zeolite molecular sieves to reduce NO_x***
    - ***Various chemical absorbers, molecular sieves & traps used to***
      - reduce HCs by converting them into H_2O & CO_2
    - ***Chemical systems are developed to trap and store H_2S and convert them into SO_2.***

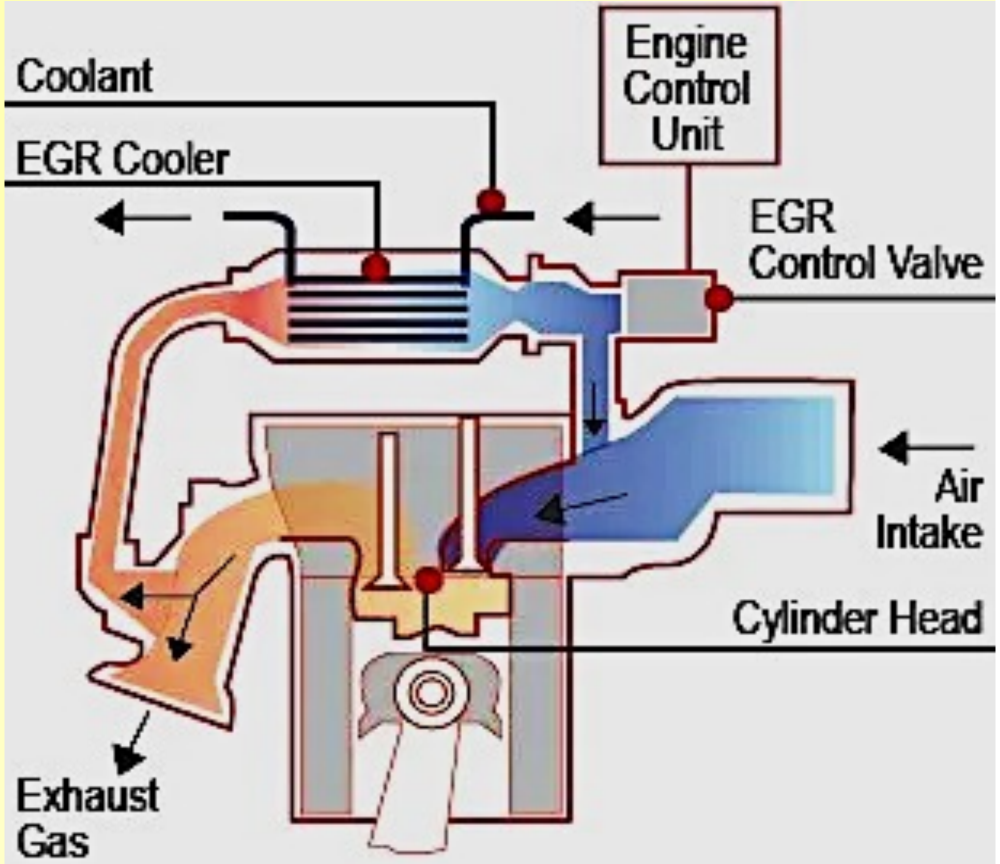


EXHAUST GAS RECIRCULATION

- ✓ NH_3 injection: in presence of catalyst, reduce NO_x, $4NH_3 + 4NO + O_2 \rightarrow N_2 + 6H_2O$ & $6NO_2 + 8NH_3 \rightarrow 7\,N_2 + 12H_2O$
- ✓ NH_3 injection not practical because of needed NH_3 storage and complex injection and control system

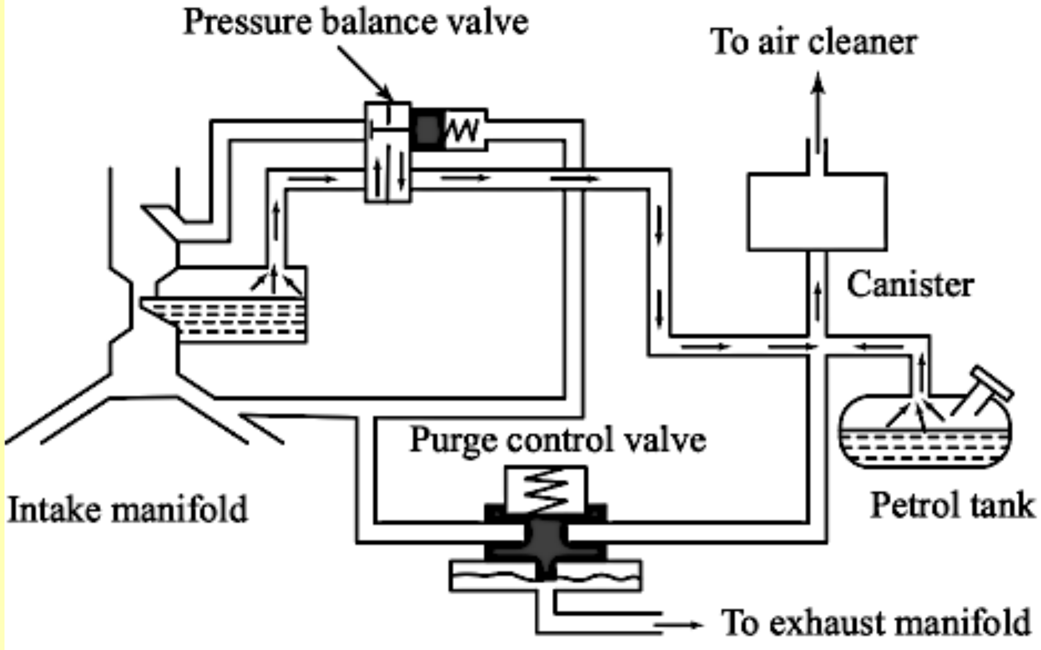
➢ ***Exhaust Gas Recirculation, EGR***

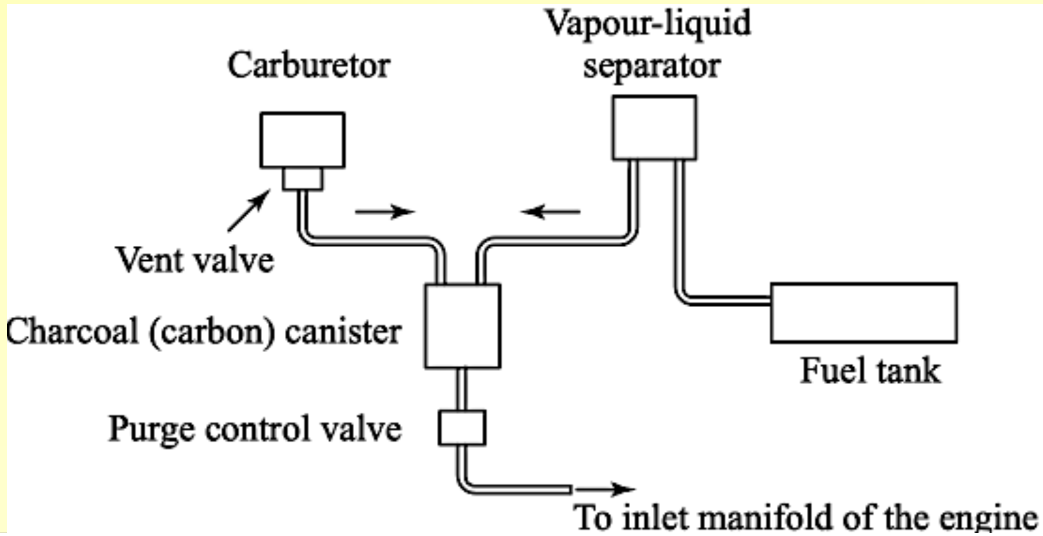
- ✓ The most effective way of reducing NO_x is to hold CC temp down by non-reacting parasite gas,
- ✓ that Absorbs energy at combustion to lower flame temp
- ✓ Adding non-reactive neutral gas in air-fuel mixture reduces flame temp and NO_x generation,
- ✓ Exhaust gas recycle (EGR) by ducting exhaust gas (15-30%) back in intake system after throttle
- ✓ No EGR at WOT (wide open throttle), idle and very little at low speeds
- ✓ EGR reduce NO_x is a costly affair of increased HC emissions and lowers thermal efficiency
- ✓ EGR increases soot, acts as abrasive & break lub, more piston-wear-piston, rings & valve trains



NON-EXHAUST EMISSION: CAUSES & CONTROL

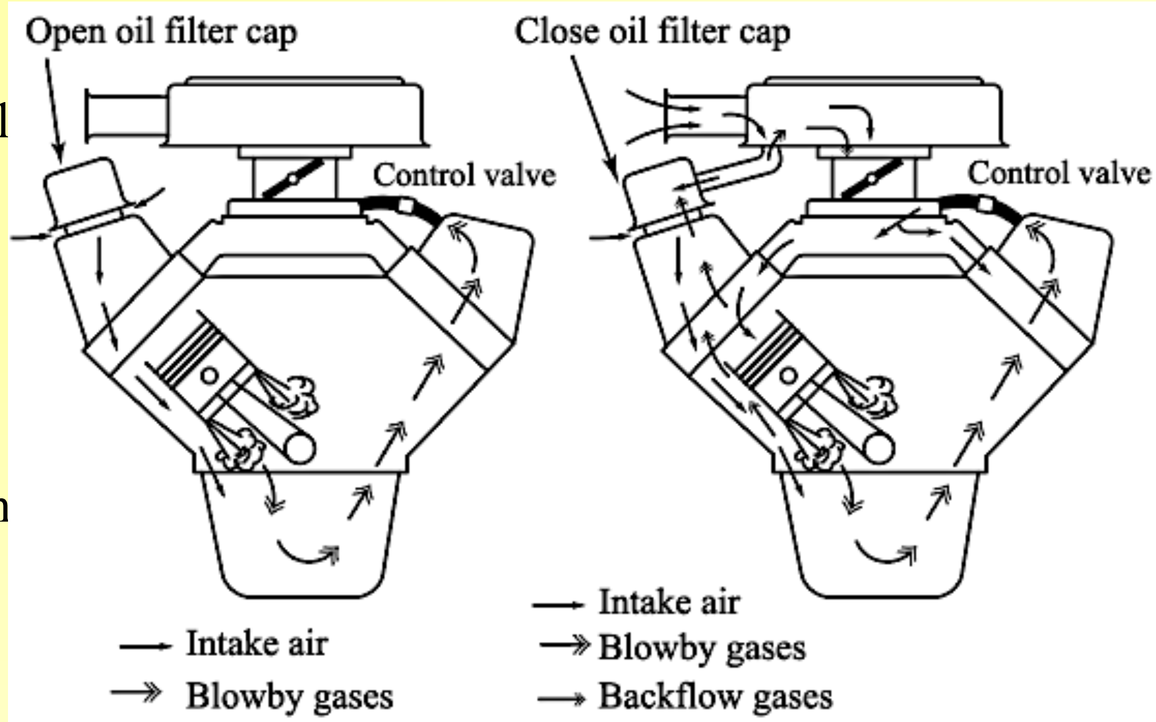
- ***Non-exhaust emissions (HCs-15 – 25%):***
- ***Evaporative emissions & Control devices***
  - ✓ 1. ***Fuel tank loss***– heating/cooling (breathing) tank emits fuel vapours via cap in atmosphere, depends on atm/tank temp, fuel composition, fuel volatility, location, amount of fuel in tank-lesser evaporation from fully filled tank (soak)
  - ✓ 2. ***Carburetor losses***: gives out fuel vapours from (i) external vent of float bowl (ii) hot soak boiling at engine stop, prevented by special devices used in evaporative controls
  - ✓ Devices to control evaporative emissions by capturing vapours in charcoal-recirculate them
  - ✓ Modern evaporative emission control system: fuel tank fitted with vapour-liquid separator, canister adsorbs fuel vapours by charcoal, at starting inlet manifold sucks fresh air through canister by purging led to intake manifold.





BLOWBY EMISSIONS & CONTROL

- ***3. Crankcase Blowby: emit gases-fuel vapours***
  - ✓ Phenomenon of charge leakage past the piston rings from cylinder to crankcase during compression / expansion stroke.
  - ✓ Blowby emission about 20 to 35% of total HCs lost from engine
  - ✓ ***Blowby control***: basic principle of positive crankcase ventilation blowby control recirculation of vapors back to intake
  - ✓ It has a tube from crankcase or rocker arm cover through a flow control valve leads into the intake manifold.
  - ✓ To provide proper ventilation to engine interiors, fresh air is drawn through rocker-arm cover opposite containing PCV system
  - ✓ The air is drawn for ventilation of blowby gases are re-routed to CC, does not allow the blowby gases to escape from system.

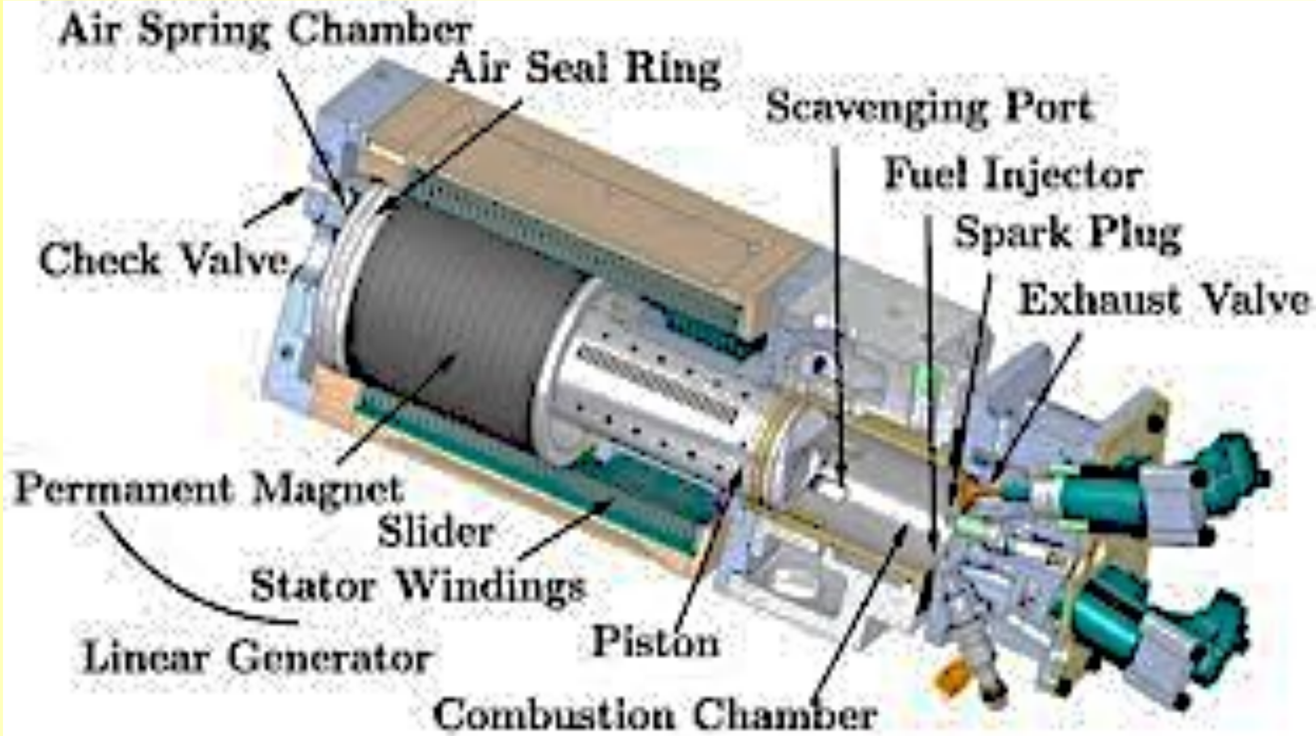


DUAL / MULTI FUEL ENGINES

- ***Many large sized stationary engines operate on dual fuels, one is Gaseous and other Liquid***
  - ✓ The dual fuel engine works on diesel and gaseous fuels, such as natural gas, sewage gas or oven gas etc. Engine can be switched over from dual fuel to single, diesel in emergency.
  - ✓ Working principle: Dual-fuel engine works on diesel cycle. The gaseous (primary) fuel is inducted in intake manifold where it mixes with air to form a homogeneous mixture.
  - ✓ The air and gas mixture is compressed similar to air in a normal diesel engine operation.
  - ✓ At end of compression stroke, a small amount of liquid fuel (5 – 7%) called pilot (secondary) fuel is injected near TDC through a conventional diesel fuel injection system
  - ✓ The pilot injection is a source of ignition, combustion starts smoothly and rapidly similar to CI engines and flame front propagates in a manner to SI engines.
  - ✓ Power output of engine is normally controlled by changing the amount of gaseous fuel.
  - ✓ Engine is capable to run on gas/diesel or a combination over wide range of mixture ratios
- ***Multi-fuel engines: can be satisfactorily operated with comparable performance and efficiency on a wide variety of fuels like diesel, crude oil, JP-4 to lighter gasoline & lubricating oils***
  - ✓ They have good idling and part-load efficiency, low exhaust smoke and noise to avoid any possible detection by the enemy.
  - ✓ The multi-fuel engines has further increased by realization of the possibilities of getting fuel economy of CI engines using gasoline.

FREE PISTON ENGINES

- ***Free piston engine is an alternative to conventional crankshaft engines, works as;***
  - ✓ 1. Optimized combustion via vcr
  - ✓ 2. Higher part load efficiency
  - ✓ 3. Possible multifuel operation
  - ✓ 4. Reduced friction losses
  - ✓ 5. Simple design
  - ✓ 6. Few moving parts

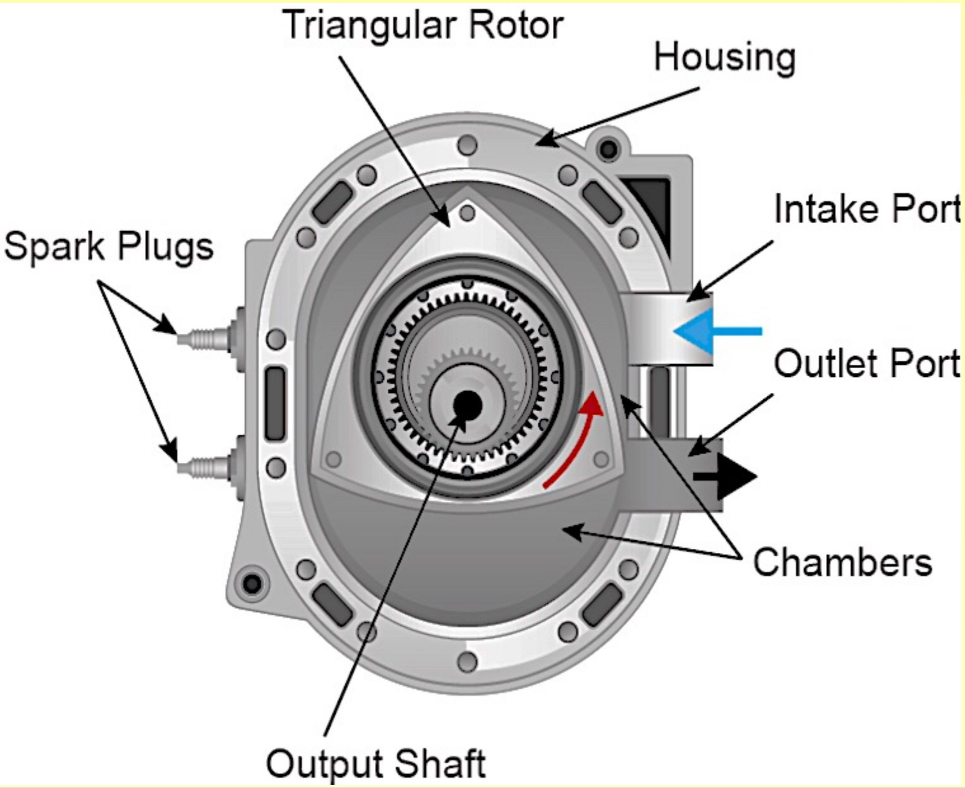


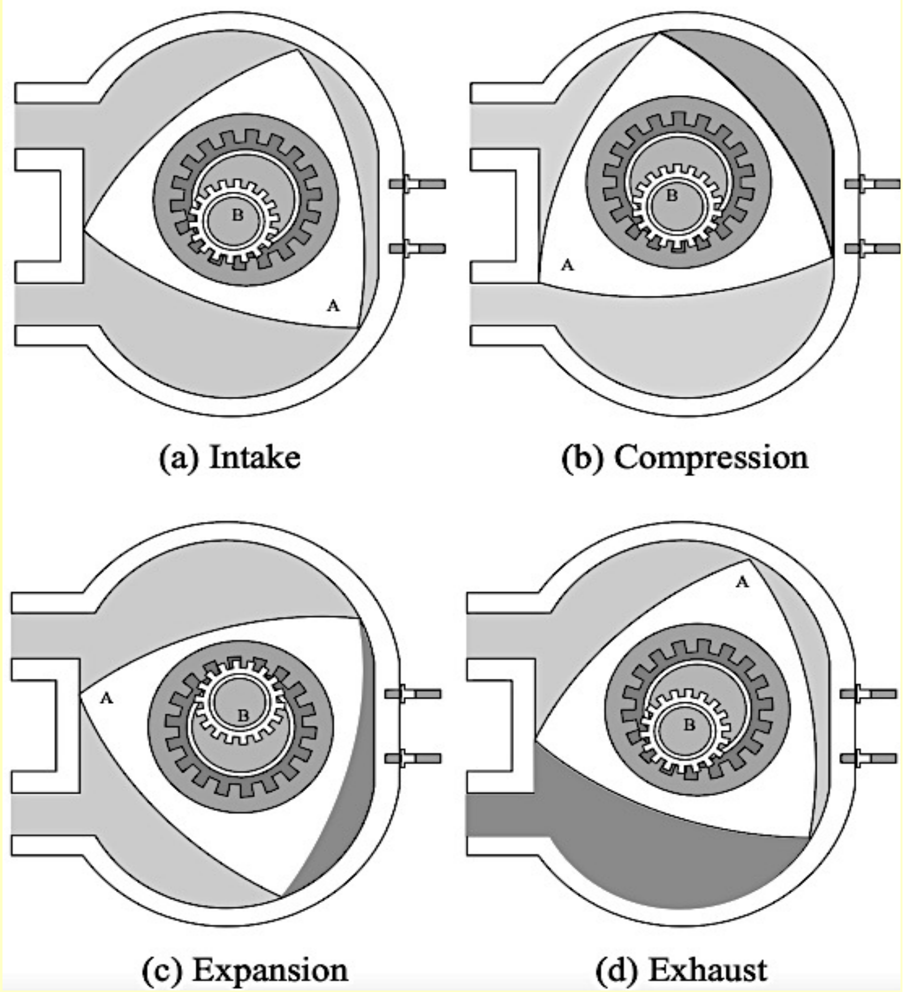
- ✓ Free piston term is used to distinguish a linear engine from a rotating crankshaft engine. The motion of piston is determined by interaction between the gas forces and load acting upon it.
- ✓ Free piston engine have characteristics as variable/length & active control of piston motion.
- ✓ Other important feature of free-piston engine are potential reductions in friction losses and possibilities to optimize engine operation using variable compression ratio.
- ✓ It consist of a combustion cylinder, a load and rebound device to store energy to compress the next cylinder charge. Hydraulic cylinder serves as both load and rebound device.

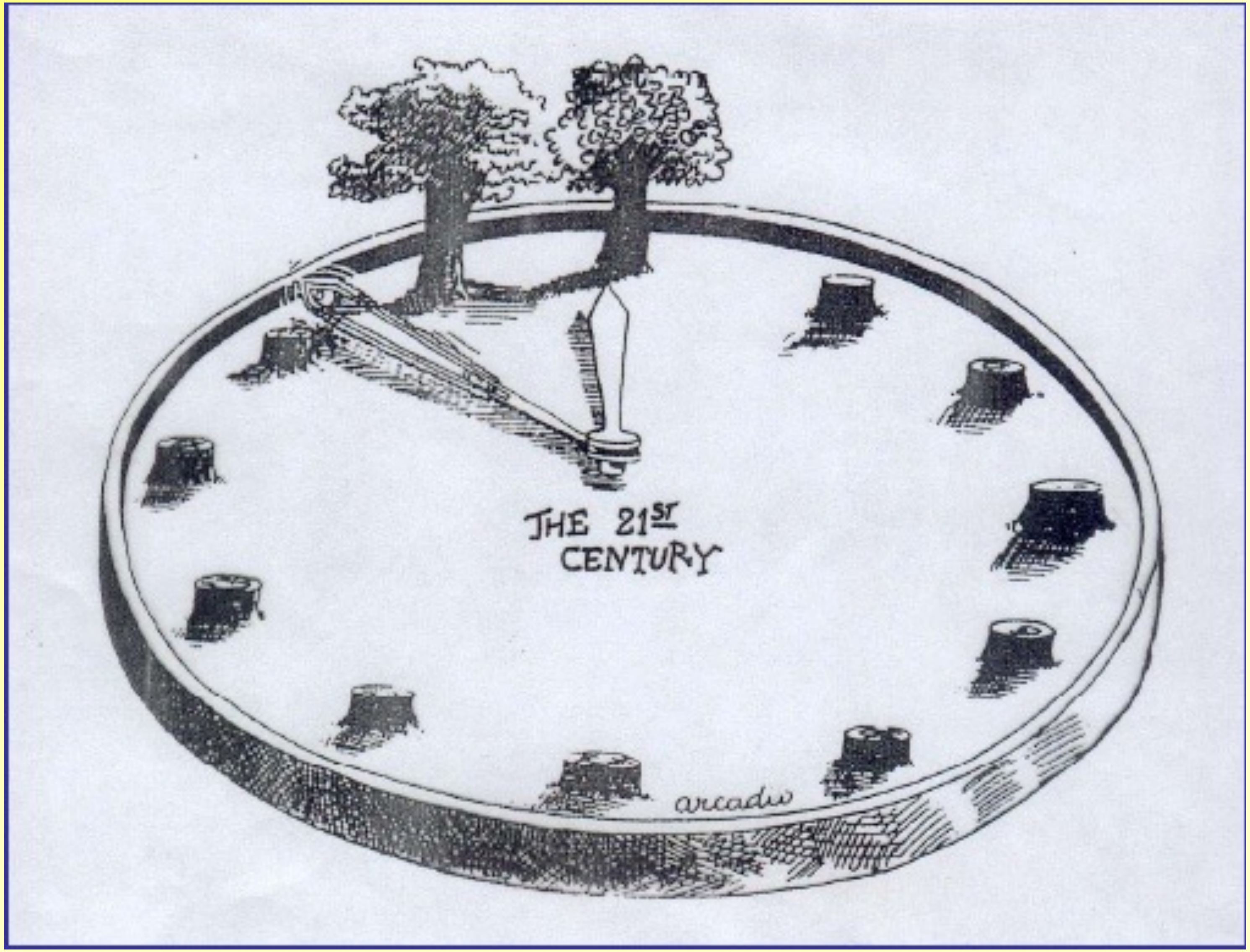
ROTARY COMBUSTION ENGINE

➢ ***Felix Wankel's Rotary Engine***

- ✓ Consists of a Δ – rotor in oval casing
- ✓ Works like a 4S engines viz. intake, compression combustion, expansion & exhaust stroke
- ✓ 3-lobed rotor on sun-planet gears form a crank-arm
- ✓ Eccentric move planet gear 3-time faster to rotor
- ✓ Compact, used in racing cars, aircraft, chain saw & auxiliary power unit, 3-power stroke per rotation







Thank You